KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2024
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates the course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1.	a)	What is meant by Natural Language Processing (NLP)? How does it differ from Machine Learning (ML)?	(07)	[CO1]
	b)	Why are NLP tasks sometimes become difficult for a machine to do? Explain with examples.	(12)	[CO2]
	c)	Differentiate between sentence-level classification and token-level classification with suitable examples.	(06)	[CO5]
	d)	Differentiate between linear regression and logistic regression. Which one is appropriate for binary text classification problem? Give a proper justification.	(10)	[CO2]
2.	a)	How does pre-training is related to fine-tuning? Explain.	(07)	[CO1]
	b)	"TF-IDF has better feature representation than BOW."– Is it true? If so, justify.	(08)	[CO3]
	c)	"FFNN is not suitable for processing sequential data." – Explain.	(08)	[CO2]
	d)	Suppose, you need to solve a 3-class text classification problem. How do you perform the task deploying Bi-LSTM? Explain with proper figure.	(12)	[CO5]
3.	a)	Draw vanilla transformer architecture and explain its main parts.	(06)	[CO1]
	b)	What are the main functions of query, key, and value vectors in transformer architecture?	(06)	[CO1]
	c)	Discuss pre-training and fine-tuning procedure of BERT model.	(15)	[CO5]
	d)	"BERT can be used as a generalized understanding model." – Explain.	(08)	[CO1]
4.	a)	How does BERT is utilized to extract features from the text? Explain with detailed diagram.	(12)	[CO5]
	b)	What is sentiment analysis? How can BERT be used to deal with the sentiment analysis task?	(08)	[CO1]
	c)	Discuss machine translation by vanilla transformer model.	(08)	[CO2]
	d)	What is co-reference resolution? Explain with example(s).	(07)	[CO1]

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5.	a)	Define corpus. What are the properties of a corpus?	(05)	[CO1]
	b)	"Patterns are greedy." – Why? Design a regular expression to answer the following statement: "Find a computer with at least 6 GHz and 16 GB RAM within 800$."	(10)	[CO1]
	c)	You have two strings "brief" and "drivers". Figure out whether "drive" is closer to "brief" or "drivers" using minimum edit distance algorithm.	(14)	[CO2]
	d)	Define stop words. What are the applications of stop word detection in NLP tasks?	(06)	[CO1]
6.	a)	Define N-gram language model. How can you compute the probability of a sentence of n words P(w), $w = w_1 w_2 \ldots\ldots w_n$? How can you calculate the maximum likelihood by bigram probabilities?	(10)	[CO1]
	b)	Consider the following corpus shown in the following figure. What is the most probable next word?	(15)	[CO3]

<s> I like Henry </s>
<s> I like college </s>
<s> Do Henry like college </s>
<s> Henry I am </s>
<s> Do I like Henry </s>
<s> Do I like college </s>

i) "<s> Do __" applying bigram probability model?
ii) "<s> Do I like __" applying bigram probability model?
iii) "<s> Do I like __" applying trigram probability model?

c) Define smoothing. How can you smooth the language model by interpolation? How to choose the value of 'λ' to maximize the probability? (10) [CO3]

7. a) What is the difference between tokens and embeddings? Explain the pipeline of creating embedding model. (08) [CO4]

b) Consider the following grammar G shown in the following figure. Is the string "The flight includes a meal." in L(G)? Answer it using CYK algorithm. (15) [CO3]

S → NP VP
NP → Det N
VP → V NP
V → includes
Det → the | a
N → meal | flight

c) Given a sequence of ice-cream observations (1 1 3) and an HMM λ = (A,B) in the following figure. Find the best hidden weather sequence Q (like H H H). (12) [CO2]

A		B	
start → HOT	0.25	P(1|HOT) =0.20 P(2|HOT) = 0.35 P(3|HOT) = 0.45	P(1|COLD) =0.50 P(2|COLD) = 0.40 P(3|COLD) = 0.10
start → COLD	0.75		
HOT → HOT	0.60		
HOT → COLD	0.40		
COLD → COLD	0.55		
COLD → HOT	0.45		

8. a) "POS tagging is a disambiguation task" – Justify the statement with example. (10) [CO1]

b) Consider the sentence: "It/PRP would/MD be/VB better/JJR to/TO book/? a/DT ticket/NN via/IN internet/NNP." The word "book" is often used as VB or NN. Given the probabilities below, find the right POS tag for the word "book". (12) [CO3]

P(VB|TO) = 0.67, P(NN|TO) = 0.02, P(book|VB) = 0.50,
P(book|NN) = 0.50, P(DT|VB) = 0.02, P(DT|NN) = 0.06.

c) For Hidden Markov Model (HMM) POS tagging, using the following formula, find the equation of calculating tag transition probabilities. (13) [CO2]

$$\hat{t}_1^n = \underset{t_1^n}{argmax}\, P(t_1^n | w_1^n)$$

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